David Moore

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Staff Software Engineer with 20+ years of Python 3 backend and AWS cloud platform experience, specializing in security-by-design systems, SOC 2 compliance programs, developer experience tooling, and operational excellence. Currently driving platform security automation and infrastructure-as-code at Optimal Dynamics, following nearly 7 years as a proven hands on technical leader at NewtonX, with a track record of building reliable, observable, and maintainable infrastructure while mentoring engineers and driving platform strategy.

SUMMARY OF ACCOMPLISHMENTS AND CORE STRENGTHS

- **Security & Compliance:** Owned NewtonX's entire cybersecurity program — maintained SOC 2 compliance through 3 consecutive annual audits and ensured ongoing GDPR adherence. Built the company's zero trust solution (Cloudflare + Sophos) from scratch, reducing non-compliant devices from hundreds to under 40. Migrated all systems to SSO & OIDC, enforced security training, and championed security-by-design across the engineering org.
- **Python 3 & AWS Platform Engineering:** Deep Python 3 expertise (Django, Celery patterns) across high-throughput production systems. AWS experience spanning compute, storage, queuing, and database tiers. Proficient with Terraform for infrastructure-as-code, Kubernetes and Helm for container orchestration, and CI/CD pipelines (Jenkins, Cloud Build, CodePipeline-equivalent patterns).
- **Operational Excellence & Observability:** Designed the engineering team's gold-standard on-call and incident response process from scratch. Reduced spurious auto-open incidents by 90% through a systematic operational excellence initiative. Championed monitoring, alerting, and observability as first-class requirements for every platform service shipped, including OpenTelemetry tracing and Grafana dashboards at Optimal Dynamics.
- **Developer Experience & Tooling:** Improved CI/CD pipelines by generating Helm charts for Kubernetes deployments, enforcing unit testing standards in pull requests, and helping migrate hand-built cloud projects to Terraform. Built internal tooling that accelerated developer workflows. Led migration from self-hosted infrastructure (Jenkins, RabbitMQ, Elasticsearch) to managed SaaS equivalents to reduce toil.
- **Event-Driven Architecture & Distributed Systems:** Designed and delivered high-throughput event-driven platforms using Kafka, RabbitMQ, and Celery — scaling async processing pipelines from 200K to 1M+ operations/day. Architected distributed worker systems for fault tolerance, horizontal scalability, and operational simplicity.
- **AI Tooling & Technical Leadership:** Led team-wide adoption of AI-assisted development tools (Anthropic Claude, GitHub Copilot) to accelerate refactoring, test generation, and prototyping. Mentored engineers at all levels; one grew to lead a sibling team. Participated in recruiting, hiring, and onboarding. Drove cross-team technical initiatives and established engineering best practices.

TECHNICAL SKILLS

Python 3 • Django • Celery • AWS (ECS/SQS/RDS/S3/DynamoDB) • GCP • Terraform • Spacelift • Bazel • Kubernetes • Helm • Docker
Kafka • RabbitMQ • Pub/Sub • Event-Driven Architecture • Distributed Systems • Microservices • REST APIs
Trino • Dagster • dbt • DuckDB • Airflow • Parquet • Preset • React • Redux • TypeScript
SOC 2 • GDPR • Zero Trust • SSO • OIDC • Auth0 • Vanta • Security Hardening • Cloudflare • Compliance Frameworks
PostgreSQL • DynamoDB • Firestore • Redis • Elasticsearch • BigTable • GitHub • Jenkins • CI/CD • Observability • OpenTelemetry • Grafana
IC Technical Leadership • Mentoring • Agile/Scrum • Incident Response • AI-Assisted Development (Claude, Copilot)

PROFESSIONAL EXPERIENCE

Optimal Dynamics, New York, NY; *Staff Software Engineer, Backend* April 2026–Present
Joined a Series C logistics/supply-chain optimization startup to strengthen platform security automation, observability, and infrastructure-as-code practices.

- Built automation to maintain SOC 2 compliance, including automated vulnerability remediation and daily public-endpoint auditing with Slack alerts. Remediated keys leaking into logs and moved secrets to SSM, strengthening security posture
- Built an agent-driven pipeline that turns Vanta findings into fix pull requests, clearing at least 45 CVEs, with two CVSS 10.0 findings fixed the day they were opened. Delivered hard upgrades required by CVEs (DuckDB 0.10 to 1.5, Dagster). These upgrades involved breaking API changes, query modifications, and performance and memory testing.
- Evaluated four observability vendors by running them, then shipped OpenTelemetry tracing on AWS X-Ray and Terraform-managed Grafana dashboards, avoiding a commercial platform quoted at about $100K a year.
- Identified and decommissioned unused AWS infrastructure (RDS, compute) left running after completed migrations, cutting cloud spend by about $22K/month (~$265K/year).
- Cut Spacelift run preparation from about 4 minutes to about 5 seconds (sampled runs) with sparse checkout and partial clones on 367 of 371 stacks; fixed stacks that could not plan, one broken since July 2025.
- Designed and built the parquet-based geographic dataset and pipeline, with blocking data-quality checks and a freshness sensor.

NewtonX, Inc., New York, NY; *Senior Staff Software Engineer / Senior Engineering Manager* October 2019–April 2026
Took full technical ownership of platform infrastructure, security, and developer experience at a high-growth B2B SaaS startup. Designed and operated Python-based cloud-native systems on GCP and AWS, ran the company's compliance program, and established the engineering practices that powered the platform's reliability and growth.

Security, Compliance & Zero Trust

- Owned the company's end-to-end cybersecurity program: maintained SOC 2 compliance through 3 consecutive annual audits, ensured GDPR adherence, and enforced security training and background check policies across the organization.
- Built the company's zero trust solution from scratch (Cloudflare + Sophos), reducing non-compliant devices from hundreds to under 40. Designed and implemented the device compliance verification system and a scalable remediation workflow for out-of-compliance employees and contractors.
- Migrated all internal systems from per-service authentication to centralized SSO and OIDC, ensuring consistent password and session policies across the org. Integrated Auth0 and Google Identity Platform; architected and maintained GCP IAM policy.
- Added automatic fraud detection to the CRM platform, reducing fraudulent activity and improving the bottom line by $75K. Led static application security testing (SAST) integration into CI/CD pipelines.
- Built the GDPR subject access request process, reducing handling time from hours to minutes.

Platform Infrastructure, Terraform & CI/CD

- Enhanced CI/CD by generating Helm charts for Kubernetes deployments, enforcing strict unit testing standards in pull requests, and helping convert hand-built GCP cloud projects to Terraform for repeatable, auditable infrastructure.
- Led migration from self-hosted infrastructure to managed SaaS: Jenkins → Google Cloud Build, RabbitMQ → managed AMQP, self-hosted Elasticsearch → Elastic.co — each planned for correctness, rollback safety, and zero downtime.
- Optimized cloud infrastructure spend by rightsizing workloads, tuning SQL databases to handle increased load with fewer resources ($1K/month saved), and eliminating unnecessary cross-zone network egress costs.

Operational Excellence & Observability

- Designed and rolled out the engineering team's on-call and incident response process from scratch — became the org's "gold standard." Reduced spurious auto-open incidents by 90% through a systematic operational excellence initiative identifying and eliminating the top recurring pain points.
- Championed observability, monitoring, and alerting as first-class requirements for all platform work — not afterthoughts. Established standards for logging, metrics, and deployment hygiene as part of the definition of done.

Distributed Systems & Event-Driven Architecture

- Designed and led delivery of a high-throughput Python 3/Django/Celery email processing platform, scaling from 200K to 1M+ operations/day using RabbitMQ for async worker queuing and Dataflow for batch processing.
- Architected a distributed worker system for high-volume asynchronous workloads, designed for horizontal scalability, fault isolation, and operational visibility.
- Led the rebuild of the core CRM platform (replacing a legacy Java monolith that reached its scaling limit) — scoped the initiative, led architecture reviews, and oversaw delivery across all modules.

AI Tooling & Developer Productivity

- Led team-wide adoption of Anthropic Claude and AI-assisted development tools to accelerate code refactoring, automated test generation, and rapid prototyping. Measurably improved delivery velocity and code quality.
- Mentored engineers at multiple levels; one grew to lead a sibling team. Participated in recruiting, interviewing, and onboarding. Wrote developer documentation and runbooks that de-tribalized platform knowledge and reduced onboarding time.

Squarespace, New York, NY; *Senior Software Development Engineer* April 2019–September 2019

- Contributed to a new microservices-based billing platform. Identified and resolved a critical Kafka serialization defect corrupting billing events in production. Participated in on-call, mentored engineers, and authored system onboarding documentation.

Amazon, New York, NY; *Software Development Engineer* September 2017–February 2019

Helped redesign the distributed architecture for scheduling and configuring content on Amazon Prime Video after the legacy system reached its scaling limit.

- Built Python migration tooling to extract and reformat data from the legacy scheduling system; integrated with DynamoDB for the new architecture.
- Increased Agile rigor and mentored new hires on system design and delivery best practices.

CoreLogic, Knoxville, TN; *Team Lead / Software Development Engineer* April 2015–August 2017

Promoted to team lead for strong backend engineering and production ownership.

- Rebuilt the central code repository server on AWS; scripted full repository migration to GitHub. Resolved critical production defects with regression test coverage.
- Improved application performance by designing a new cache architecture with verified throughput gains. Led security auditing for widely deployed MLS software.

Courion, Westborough, MA; *Staff Software Engineer / Senior Security Software Engineer / Design Lead* December 2005–April 2015

Won 2 President's Club awards (rarely given to engineers) for software delivery and security leadership during the company's migration from on-premises to cloud-hosted SaaS.

- Designed and built identity access management software: permission management, user provisioning/deprovisioning, password reset, and self-service account unlock.
- Led architecture of cryptography and hashing upgrades; owned the full security fix lifecycle from design through test execution and release.

MAPASS, Inc.; *Software Engineer (Founding Engineer)* December 2002–April 2005

- First engineer at an early-stage startup; architected and built the full product from scratch, including third-party integrations, email automation, and a customer-facing web portal.

EDUCATION

B.S., Computer Science — University of Illinois at Urbana-Champaign, IL (May 2001)